

Air Quality Insights Dashboard: Defining a Product Opportunity from Connected Vehicle Data

Overview

A large western state Department of Transportation (DOT) had secured grant funding for connected vehicle initiatives but had not yet determined which opportunity area would deliver the greatest value. Multiple opportunity areas were under consideration, yet none had been sufficiently defined to justify product investment. Working alongside researchers, data engineers, hardware engineers, customer stakeholders, and executive leadership, I led the research effort to evaluate competing concepts, identify customer needs, and determine which opportunity should move forward into product development. The resulting work led to selection of Air Quality Insights as the preferred investment area, informed development of a customer-facing analytics dashboard, secured ~\$200,000 in contracted work, and was later expanded into an industry publication accepted for presentation at the Intelligent Transport Systems World Congress in Dubai, UAE.

My Role: Lead Researcher & Product Measurement Lead

I was responsible for:

- Discovery research design and execution
- Stakeholder surveys and interviews
- Opportunity prioritization
- Co-authoring the customer proposal that secured project funding
- Focus group facilitation
- Product measurement strategy
- Telemetry and metric evaluation
- Dashboard requirements definition
- Executive presentations and customer readouts

Discovery Research: Identifying the Right Opportunity

Rather than beginning with a predetermined solution, we explored multiple applications of connected vehicle data to determine which opportunity area would provide meaningful value for transportation agencies. I conducted survey research with approximately ten stakeholders and subject matter experts, followed by six in-depth interviews with transportation leaders, engineers, and practitioners. These activities focused on understanding operational workflows, reporting needs, environmental priorities, and funding objectives. Research participants evaluated candidate use cases, reporting needs, and workflow requirements to help identify where connected vehicle data could create the greatest value.

A key finding emerged during this phase. Stakeholders consistently expressed greater interest in actionable operational insights than raw telemetry data. While environmental reporting was viewed as valuable, participants repeatedly connected emissions-related measures to a broader challenge: understanding and reducing traffic congestion. This insight elevated Air Quality Insights above competing concepts and established a clear direction for further validation.

Concept Validation & Product Definition

With Air Quality Insights identified as the leading opportunity, I led a second phase of research focused on validating concepts, refining metrics, and understanding how insights should be delivered to end

users. I facilitated focus groups with representatives from state transportation agencies, academic researchers, and private-sector transportation experts. These sessions explored reporting needs, workflow integration, metric prioritization, and reporting concepts.



Figure 2. Concept validation workshop collateral used to validate reporting concepts, workflow requirements, metric priorities

This phase revealed that emissions hotspots served a dual purpose. In addition to supporting environmental reporting objectives, they acted as a practical proxy for traffic congestion, helping agencies identify operational inefficiencies and prioritize roadway improvements.

The research directly informed:

- Dashboard structure and information architecture
- Metric selection and prioritization
- Reporting requirements
- Segmentation capabilities

From Research to Product

Research findings were translated into a customer-facing Air Quality Insights Dashboard that enabled transportation agencies to monitor congestion and emissions trends using connected vehicle telemetry. Throughout development, I partnered closely with engineering teams to evaluate telemetry quality, validate metrics, assess hardware performance, and ensure that the resulting product aligned with customer workflows and decision-making needs.

The final solution transformed large-scale connected vehicle data into actionable operational insights, allowing agencies to identify congestion hotspots, evaluate intervention opportunities, and support environmental planning initiatives.

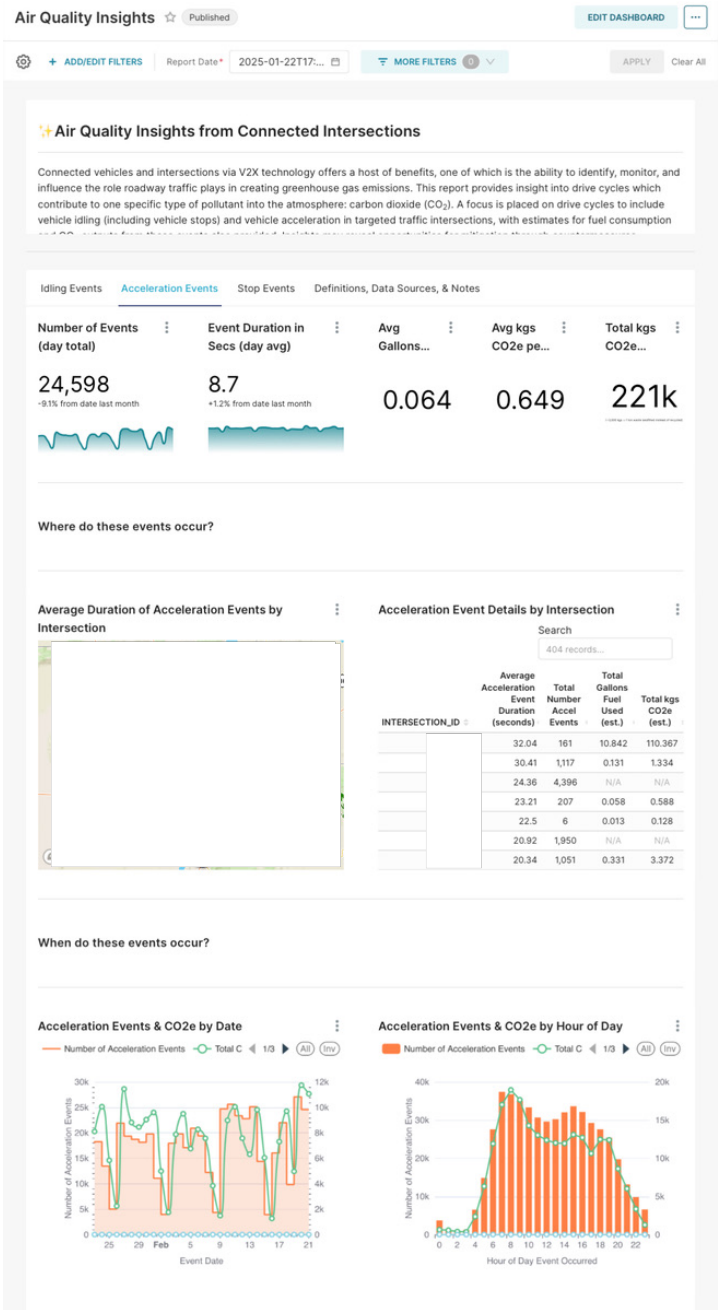


Figure 3. Final Air Quality Insights Dashboard delivered to customer stakeholder (identifying information redacted).

Business Impact

The project transformed a loosely defined grant opportunity into a funded product initiative. I co-authored the customer proposal that secured ~\$200k in contracted work and coordinated delivery across multiple teams to ensure successful execution within SoW commitments and project timelines. The customer accepted the final deliverables, funded continued development efforts, and incorporated the findings into ongoing transportation and environmental planning activities.

Beyond the customer engagement itself, the foundational effort was later expanded into an industry publication: **Leveraging V2X Data Insights on Air Quality to Secure Funding**. The paper was accepted for presentation at the Intelligent Transport Systems World Congress in Dubai, UAE, extending the impact of the research beyond a single customer engagement and into the broader industry.

Key Outcomes

- Identified and validated the highest-value opportunity, resulting in selection of Air Quality Insights as a funded product initiative
- Defined a new product direction supported by customer demand and operational value
- Informed dashboard design, metric selection, and MVP requirements
- Secured approximately \$200,000 in customer-funded work
- Delivered a customer-facing analytics product adopted by stakeholders
- Published industry thought leadership accepted for international presentation